EXPERIMENT – 6

Snort IDS/IPS to Detect and Prevent Real-Time Threats in TryHackMe Platform

AIM

To detect and prevent real-time network attacks using Snort IDS/IPS on the TryHackMe platform.

SOFTWARE / TOOLS REQUIRED

TryHackMe account
AttackBox or VPN connection
Snort IDS/IPS
Nmap
Linux Terminal

PROCEDURE

1. Starting TryHackMe Environment
2. Login to TryHackMe.
3. Search for the Snort room.
4. Click Start Machine.
5. Click Start AttackBox.
   Open Terminal in AttackBox.
6. Verify Snort Installation
   snort -V
7. Test Snort Configuration
   sudo snort -T -c /etc/snort/snort.conf
   EXPERIMENT 1: ICMP (Ping) Detection – IDS Mode
   Edit Local Rules
   sudo nano /etc/snort/rules/local.rules
   Add Rule
   alert icmp any any -> any any (msg:"ICMP Ping Detected"; sid:1000001; rev:1;)
   Save and Exit
   Ctrl + O → Enter
   Ctrl + X
   Run Snort in IDS Mode
   sudo snort -c /etc/snort/snort.conf -i eth0
   Generate ICMP Traffic (New Terminal Tab)
   ping 8.8.8.8
   Observation
   Alert “ICMP Ping Detected” is displayed in Snort terminal.

   (Optional log check)
   sudo tail -f /var/log/snort/alert

EXPERIMENT 2: Port Scan Detection
Edit Rules File
sudo nano /etc/snort/rules/local.rules
Add Rule
alert tcp any any -> any any (flags:S; msg:"Port Scan Detected"; sid:1000002; rev:1;)
Restart Snort
Ctrl + C
sudo snort -c /etc/snort/snort.conf -i eth0
Run Nmap Scan (New Terminal)
nmap -sS 8.8.8.8
Observation
Snort detects the SYN scan and displays “Port Scan Detected” alert.
EXPERIMENT 3: IPS Mode – Blocking ICMP (Inline Mode)
Modify Rules
sudo nano /etc/snort/rules/local.rules
Add Rule
drop icmp any any -> any any (msg:"ICMP Blocked"; sid:1000003; rev:1;)
Run Snort in Inline (IPS) Mode
sudo snort -Q --daq afpacket -i eth0:eth0 -c /etc/snort/snort.conf
Generate ICMP Traffic (New Terminal)
ping 8.8.8.8

(Optional)

sudo tail -f /var/log/snort/alert
NOTE

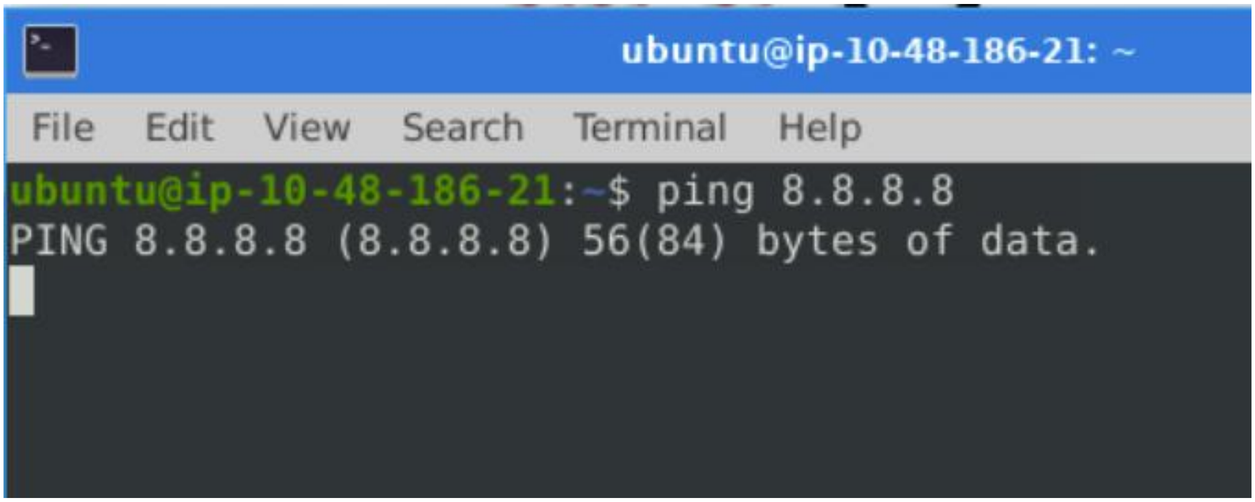
If eth0 does not work:

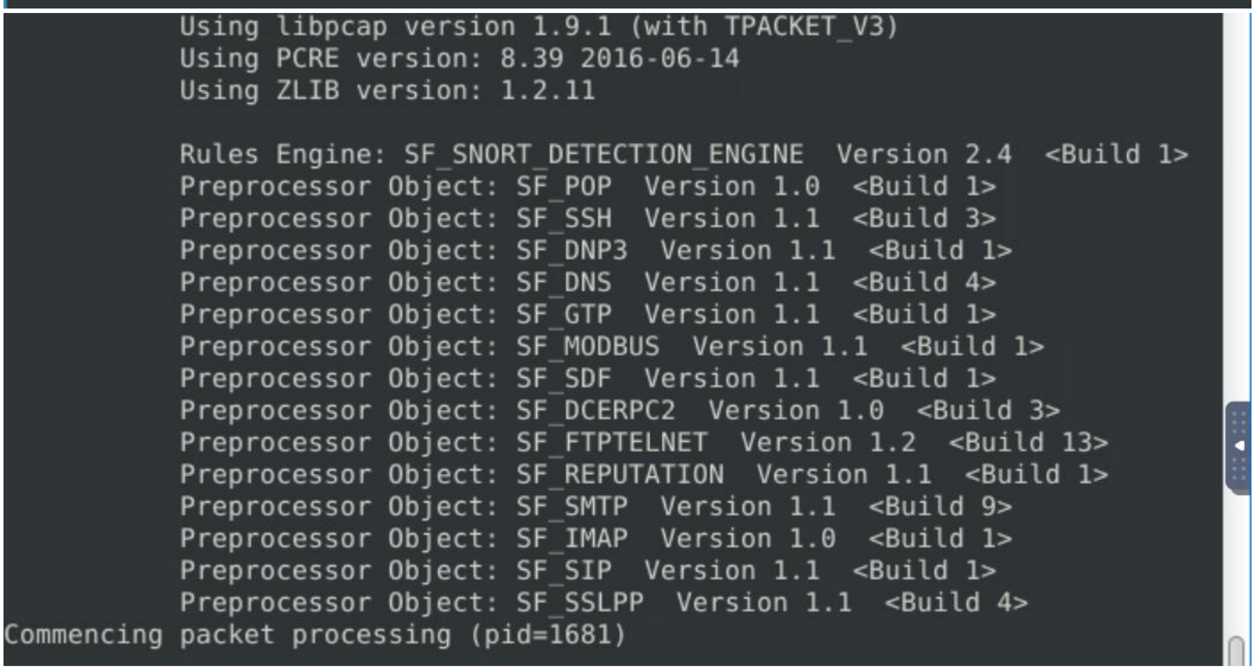
ip a

Use the correct interface name instead of eth0.

OUTPUT

Snort successfully detected ICMP traffic, identified port scanning activity, and blocked ICMP packets in inline IPS mode.





RESULT

Thus, Snort IDS/IPS was successfully configured on the TryHackMe platform to detect and prevent real-time network threats.